Appendix. Sensitivity of the wavelet packet decomposition (WPD) to the choice of wavelet basis (db4 against other commonly used bases)

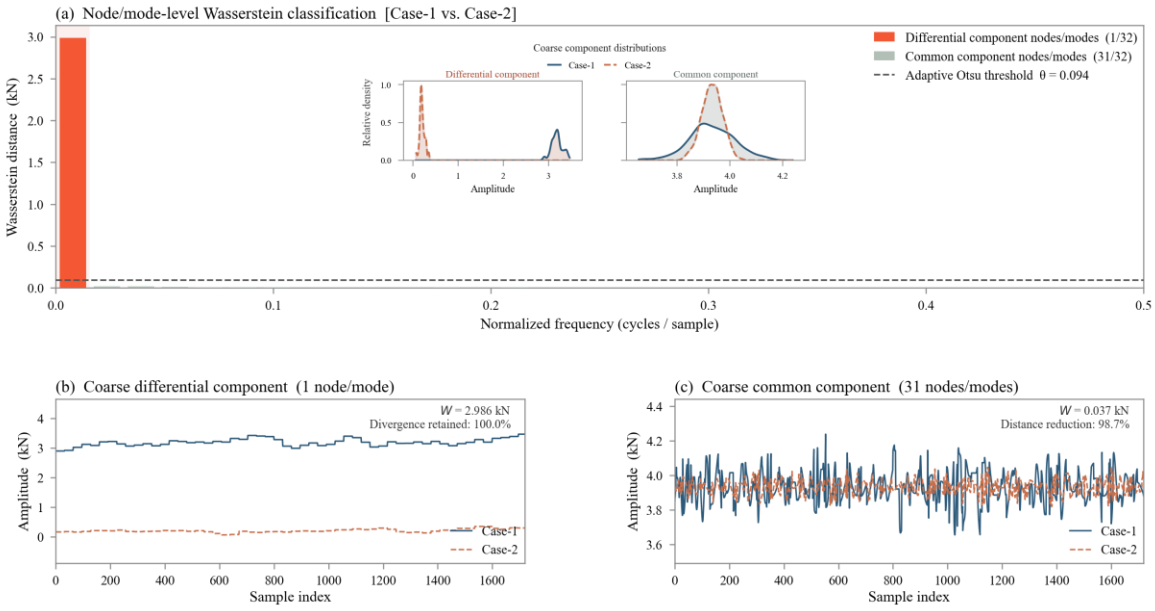
1.1 Basis for the choice and comparison design

The choice of wavelet basis for the first stage of the two-stage funnel-shaped framework is subject to one hard constraint: the wavelet must be orthogonal, so that the reconstructed leaf nodes sum exactly to the original signal. This is the prerequisite for the first-stage coarse result to enter the subsequent four-component merging. Among the orthogonal wavelets that satisfy it, the Daubechies family is compactly supported with good time-domain localization, which suits the transient content in MWD signals caused by rock breaking and drill-tool vibration; with four vanishing moments, db4 balances support length against frequency-domain selectivity and is a standard choice in mechanical-vibration signal processing. The main text accordingly adopts the Daubechies-4 (db4) wavelet.

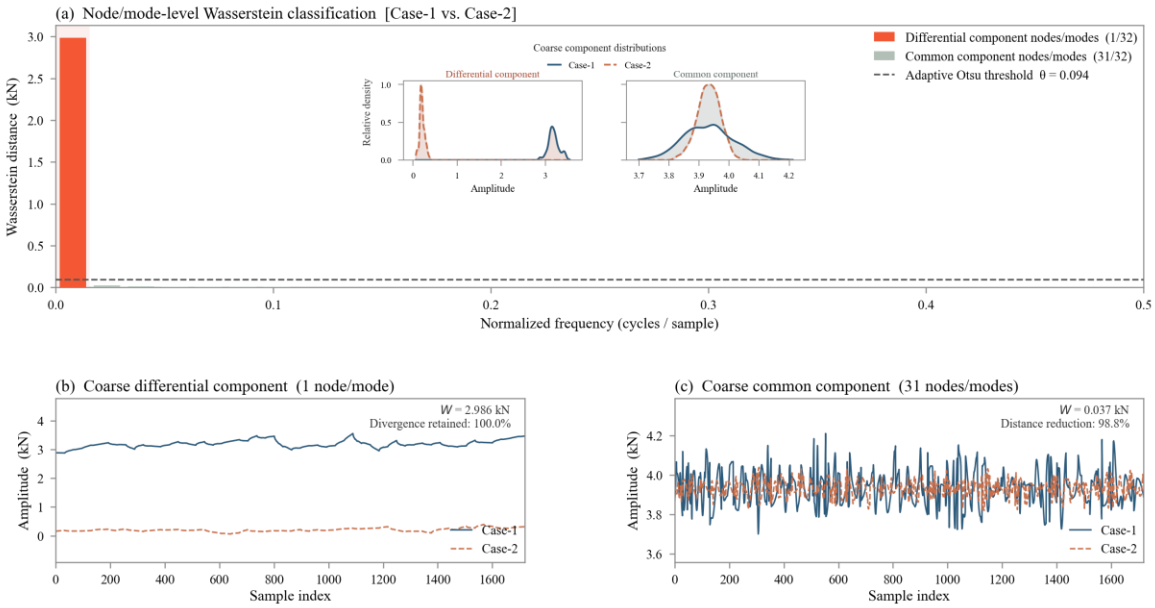
To test how much that choice actually affects the separation, this appendix compares db4 systematically with five other widely used orthogonal bases — haar, db2, db6, db8 and sym4 — spanning a representative range from one vanishing moment (haar) to eight (db8). The number of WPD levels is determined adaptively from the signal length and is independent of the wavelet basis, so all six bases yield the same number of leaf nodes at a given level. The comparison therefore holds the number of levels, the node-discrimination mechanism (node-wise Wasserstein distance with an adaptive Otsu threshold) and the second-stage refinement entirely fixed, and replaces only the wavelet basis itself. The db4 results are loaded from the decomposition data saved for the main text; the other five are recomputed with the same node-reconstruction routine.

1.2 Sensitivity of the first-stage coarse decomposition to the wavelet basis

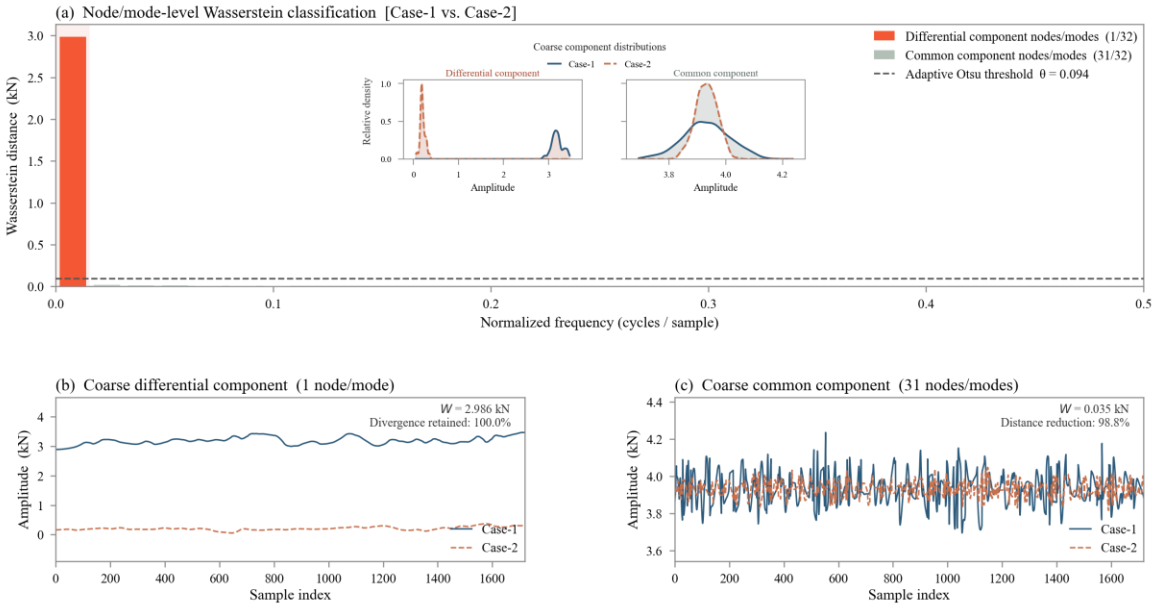
This section compares the first-stage coarse decomposition itself across the six bases. The case is again the single-factor rotational-speed pair of Sections 2.2.1 and 2.2.2 of the main text (Case-1 vs Case-2, pressure signal); results are given in Fig. 1 and Table 1.



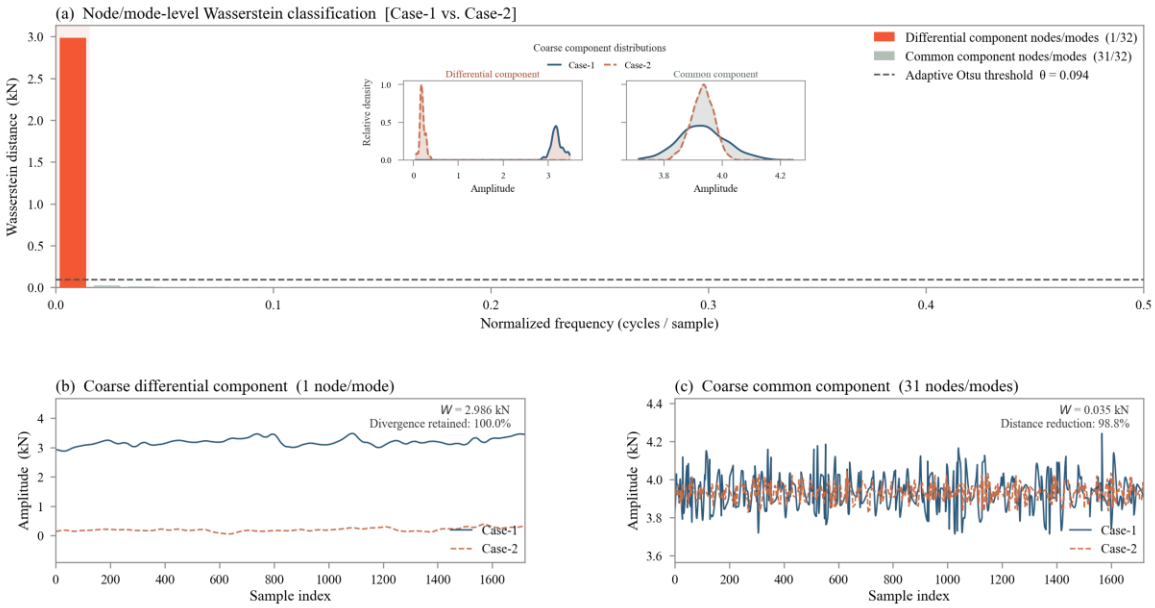
(a) haar


(a) Node/mode-level Wasserstein classification [Case-1 vs. Case-2]
Wasserstein distance (kN)
Normalized frequency (cycles / sample)
Differential component nodes/modes (1/32)
Common component nodes/modes (31/32)
Adaptive Otsu threshold θ = 0.094
Coarse component distributions
Case-1
Case-2
Differential component
Common component
Relative density
Amplitude
(b) Coarse differential component (1 node/mode)
W = 2.986 kN
Divergence retained: 100.0%
Amplitude (kN)
Sample index
(c) Coarse common component (31 nodes/modes)
W = 0.037 kN
Distance reduction: 98.8%

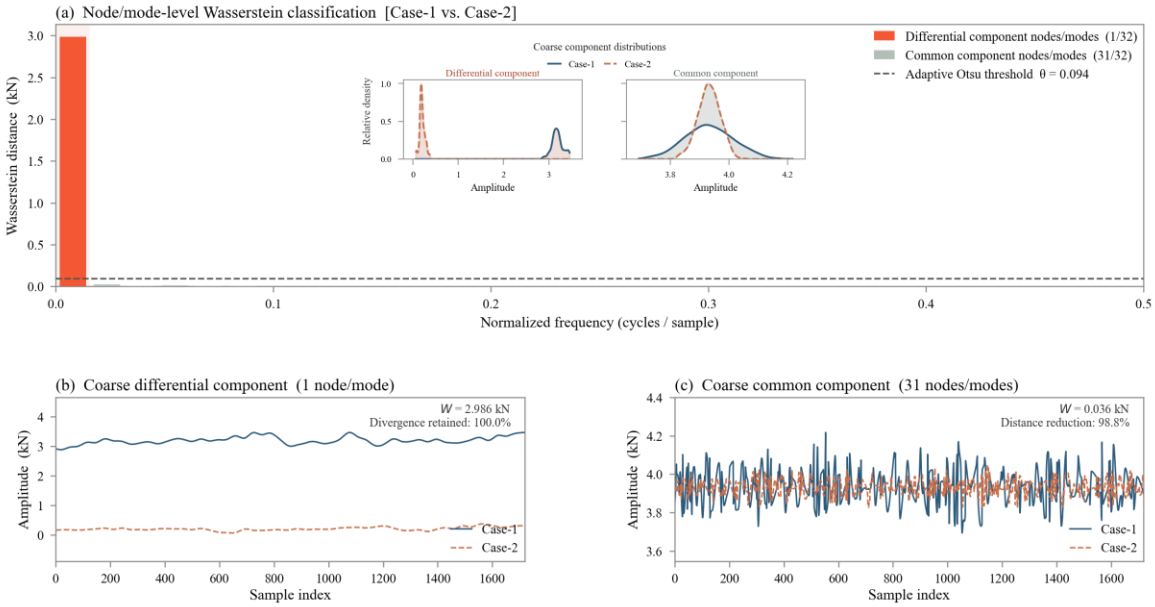
(b) db2


(a) Node/mode-level Wasserstein classification [Case-1 vs. Case-2]
Wasserstein distance (kN)
Normalized frequency (cycles / sample)
Differential component nodes/modes (1/32)
Common component nodes/modes (31/32)
Adaptive Otsu threshold θ = 0.094
Coarse component distributions
Case-1
Case-2
Differential component
Common component
Relative density
Amplitude
(b) Coarse differential component (1 node/mode)
W = 2.986 kN
Divergence retained: 100.0%
Amplitude (kN)
Sample index
(c) Coarse common component (31 nodes/modes)
W = 0.035 kN
Distance reduction: 98.8%

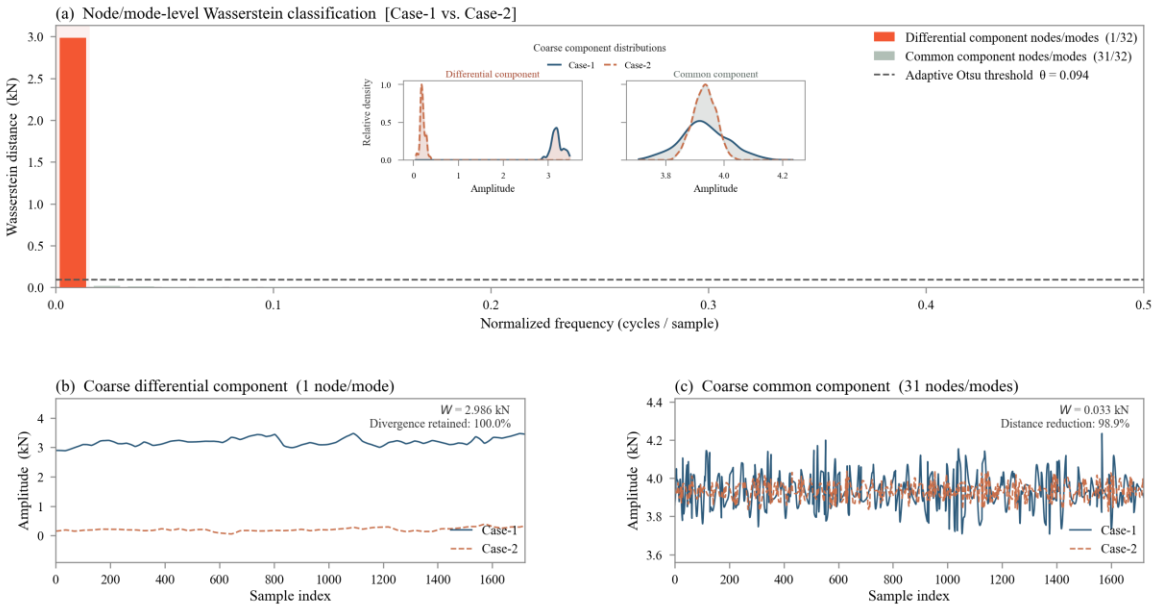
(c) db4


(a) Node/mode-level Wasserstein classification [Case-1 vs. Case-2]
Wasserstein distance (kN)
Normalized frequency (cycles / sample)
Differential component nodes/modes (1/32)
Common component nodes/modes (31/32)
Adaptive Otsu threshold θ = 0.094
Coarse component distributions
Case-1
Case-2
Differential component
Common component
Relative density
Amplitude
(b) Coarse differential component (1 node/mode)
W = 2.986 kN
Divergence retained: 100.0%
Amplitude (kN)
Sample index
(c) Coarse common component (31 nodes/modes)
W = 0.035 kN
Distance reduction: 98.8%

(d) db6


(a) Node/mode-level Wasserstein classification [Case-1 vs. Case-2]
Wasserstein distance (kN)
Normalized frequency (cycles / sample)
Differential component nodes/modes (1/32)
Common component nodes/modes (31/32)
Adaptive Otsu threshold θ = 0.094
Coarse component distributions
Case-1
Case-2
Differential component
Common component
Relative density
Amplitude
(b) Coarse differential component (1 node/mode)
W = 2.986 kN
Divergence retained: 100.0%
Amplitude (kN)
Sample index
(c) Coarse common component (31 nodes/modes)
W = 0.036 kN
Distance reduction: 98.8%

(e) db8



(f) sym4

Fig. 1. Node classification by Wasserstein distance and the resulting common- and differential-component waveforms for db4 and haar/db2/db6/db8/sym4 (Case-1 vs Case-2, pressure signal).

Table 1. Key indices for the six wavelet bases (Case-1 vs Case-2, pressure signal).

Wavelet basis	Otsu threshold	Common / differential nodes	Relative reconstruction error (A / B)	CC distance reduction	DiC distance retention
haar	0.0942	31/1	6.99e-16 / 7.54e-16	98.75%	100.00%
db2	0.0937	31/1	6.05e-16 / 6.06e-16	98.77%	99.99%
db4	0.0937	31/1	4.50e-16 / 4.70e-16	98.84%	100.00%
db6	0.0937	31/1	4.53e-16 / 4.81e-16	98.84%	100.00%
db8	0.0937	31/1	4.71e-16 / 5.05e-16	98.78%	99.99%
sym4	0.0936	31/1	5.61e-14 / 3.95e-13	98.88%	100.00%

The relative reconstruction errors of all six bases are of machine-precision order (4.5×10^{-16}–3.9×10^{-13}) and the node discrimination is identical in every case (31 common nodes and one differential node). The CC distance reduction lies between 98.75% and 98.88% and the DiC distance retention between 99.99% and 100.00%, the six bases differing by at most 0.13 percentage points, so the first-stage coarse separation is insensitive to the choice of wavelet basis.

1.3 Final separation of the complete two-stage workflow (single case)

This section applies the complete two-stage workflow to all six bases. The final separation indices are given in Table 2; the four sub-components share the same source as Fig. 1 and likewise differ negligibly among the bases, so they are not plotted individually.

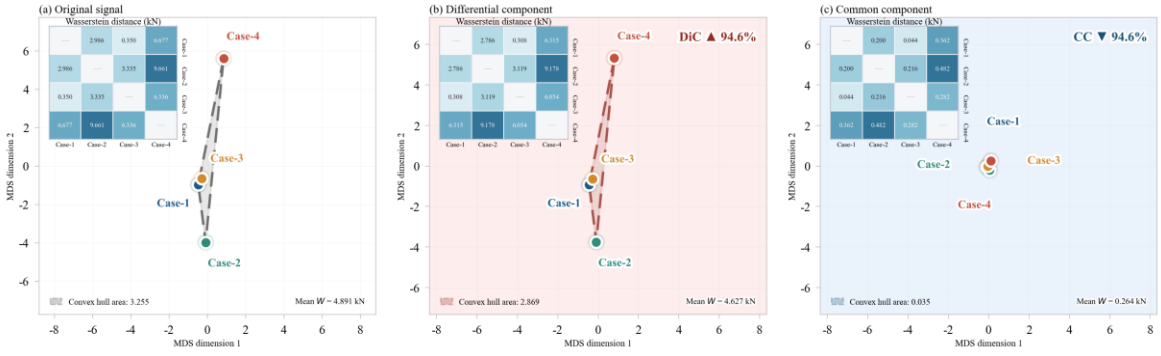
Table 2. Final DiC retention and CC reduction for the six wavelet bases (Case-1 vs Case-2, pressure signal). Percentages are given to two decimal places so that the differences among bases remain visible.

Wavelet basis	Final DiC retention	Final CC reduction	(reference: first-stage coarse components)
haar	93.31%	93.31%	Coarse DiC 100.00% / coarse CC 98.75%
db2	93.28%	93.28%	Coarse DiC 99.99% / coarse CC 98.77%
db4	93.27%	93.27%	Coarse DiC 100.00% / coarse CC 98.84%
db6	93.27%	93.27%	Coarse DiC 100.00% / coarse CC 98.84%
db8	93.26%	93.26%	Coarse DiC 99.99% / coarse CC 98.78%
sym4	93.27%	93.27%	Coarse DiC 100.00% / coarse CC 98.88%

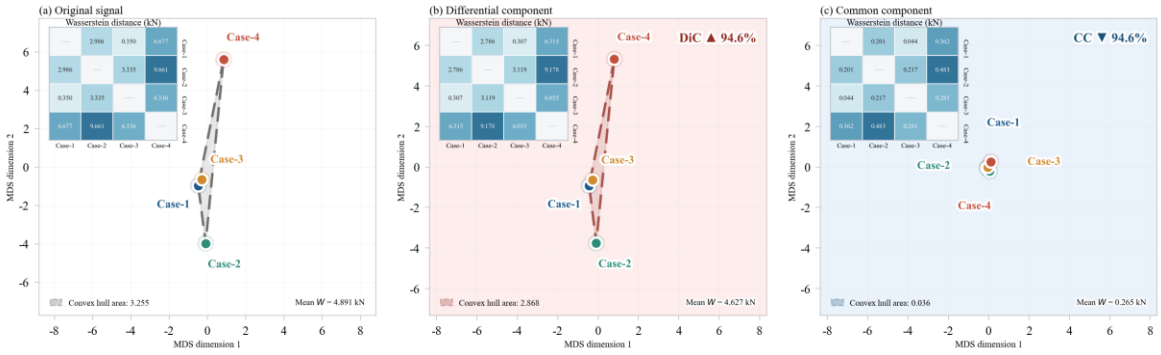
After the complete two-stage workflow, the final DiC retention and CC reduction of the six bases lie between 93.26% and 93.31%, a spread of at most 0.04 percentage points and narrower than that of the first-stage coarse components, showing that the second-stage soft-mask refinement also acts to harmonize results across wavelet bases.

1.4 Generality check over all six borehole pairs

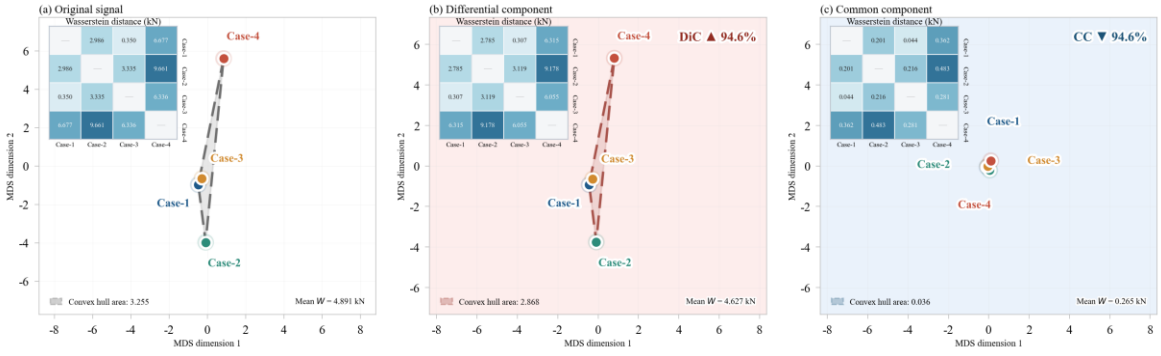
This section applies each of the six bases to all four boreholes of the main dataset of the main text (Case-1 to Case-4, six pairs in total). The original signal, differential component and common component are projected onto a plane by MDS and their convex hulls drawn, so that convergence of the common component and continued divergence of the differential component can be read directly; see Fig. 2 and Table 3.



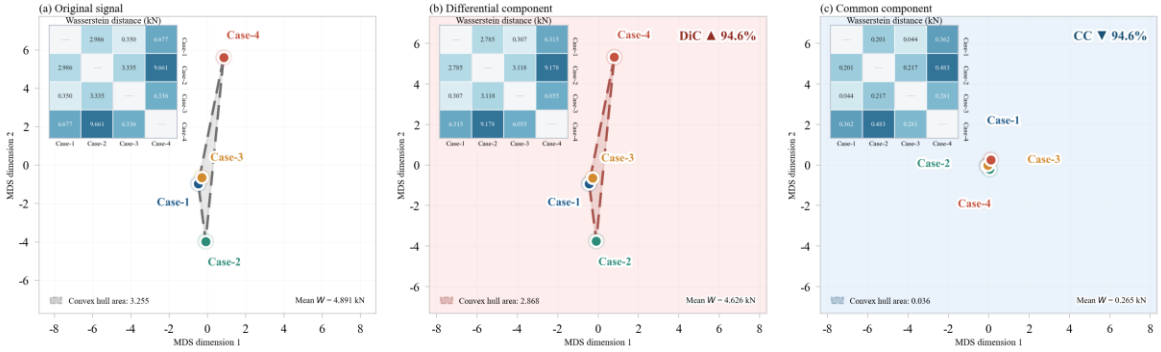
(a) haar



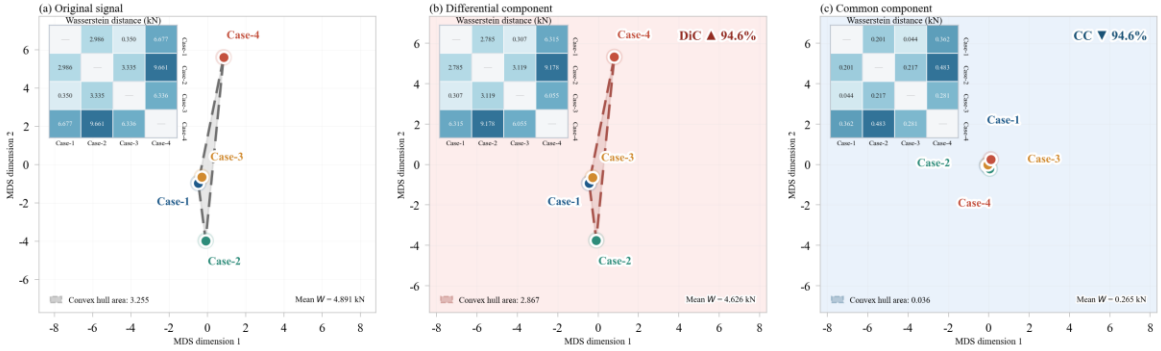
(b) db2



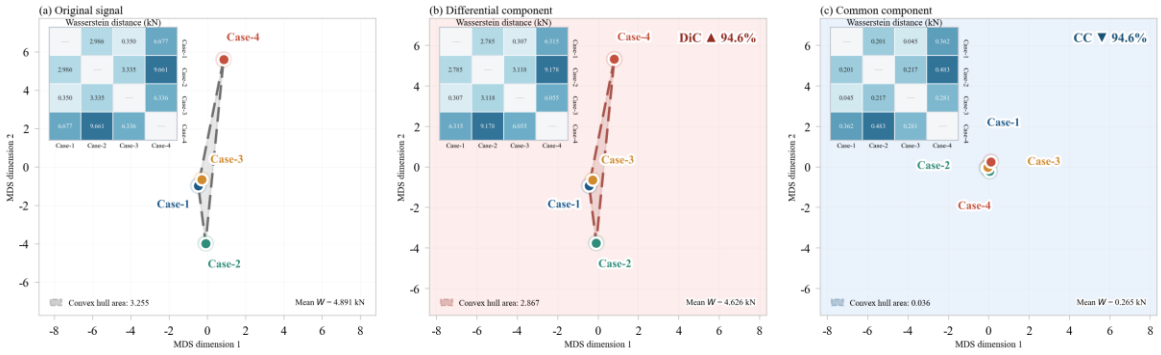
(c) db4



(d) db6



(e) db8



(f) sym4

Fig. 2. Wasserstein spatial structure over four boreholes for db4 and haar/db2/db6/db8/sym4 (Case-1 to Case-4, pressure signal, MDS projection).

Table 3. Key indices of the Wasserstein spatial structure over four boreholes (six pairs) (means, pressure signal)

Wavelet basis	CC distance reduction	DiC distance retention	CC convex hull area
haar	94.59%	94.60%	0.035
db2	94.59%	94.59%	0.036
db4	94.59%	94.59%	0.036
db6	94.59%	94.59%	0.036
db8	94.59%	94.59%	0.036
sym4	94.59%	94.59%	0.036

The six-pair means agree with the single case: the CC distance reduction is 94.59% for all six bases (differing by at most 0.01 percentage points), and their convex-hull structures for the original signal, differential component and common component (the groups of Fig. 2) are visually almost identical, further confirming that the two-stage funnel-shaped framework is robust to the choice of first-stage wavelet basis.

1.5 Conclusions

Six representative orthogonal wavelet bases, from haar with one vanishing moment to db8 with eight, already perform almost identically at the first-stage coarse separation (CC distance reduction 98.75%–98.88%) and give identical node discrimination (31 common nodes and one differential node in every case). After the complete two-stage workflow the spread narrows further (93.26%–93.31%), and over all six borehole pairs the CC distance reduction is 94.59% for every basis, differing by at most 0.01 percentage points.

The separation performance of the two-stage funnel-shaped framework therefore does not depend on a fine choice among wavelet bases: provided the hard constraint of orthogonality is met, the node discrimination and the subsequent refinement yield consistent results across bases. The choice of db4 in the main text rests on orthogonality (a structural prerequisite of the framework) and compact support (suited to the transient content of MWD signals), and the results here further show that this choice is not a sensitive hyperparameter of the method.